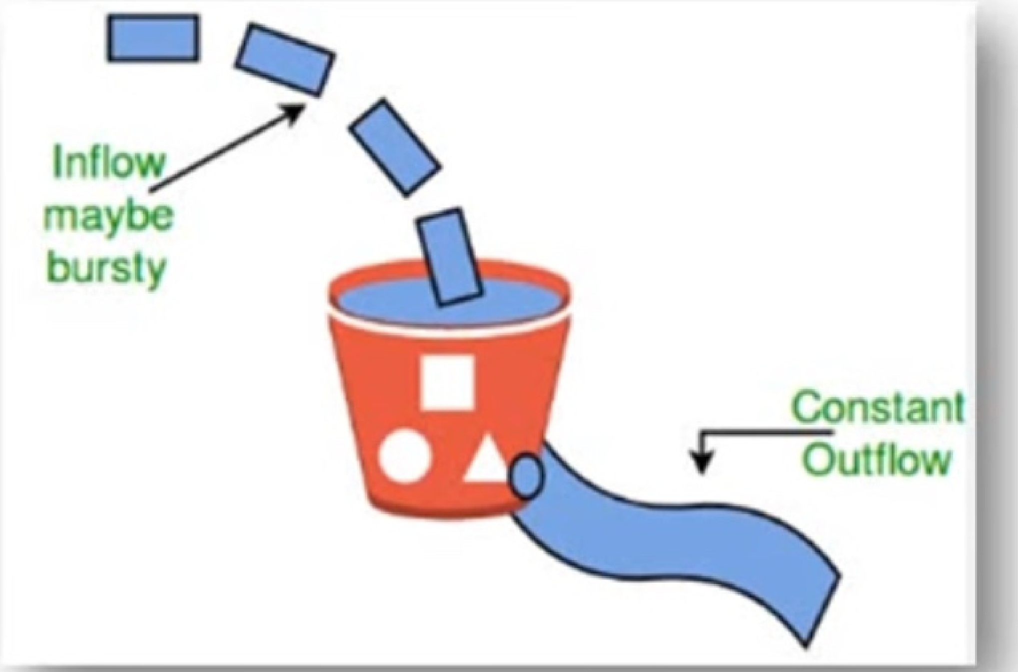
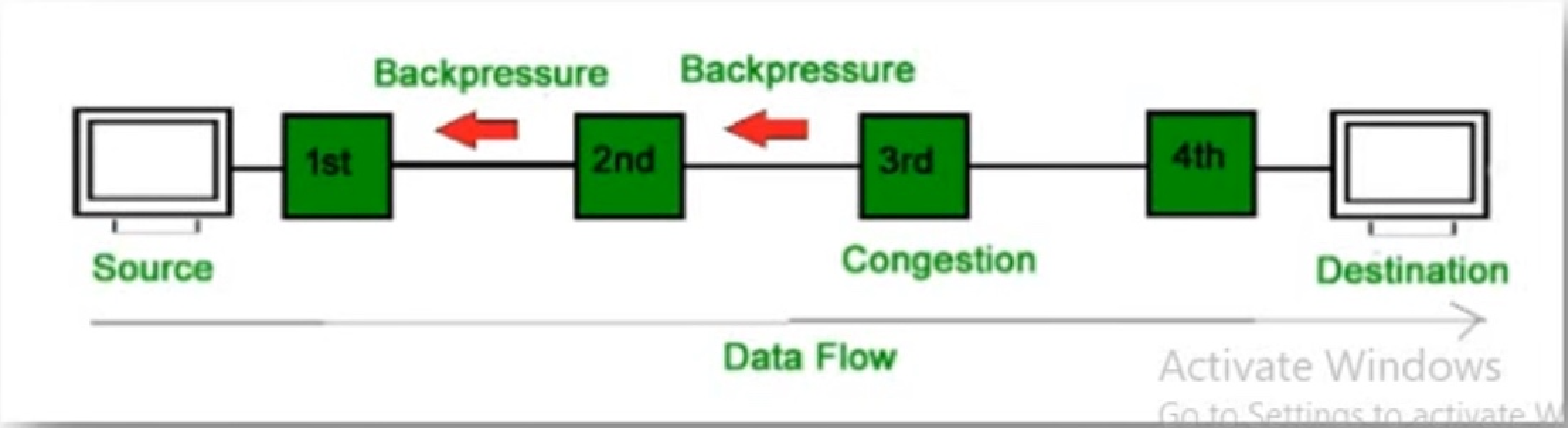
Congestion Control in Transport layer

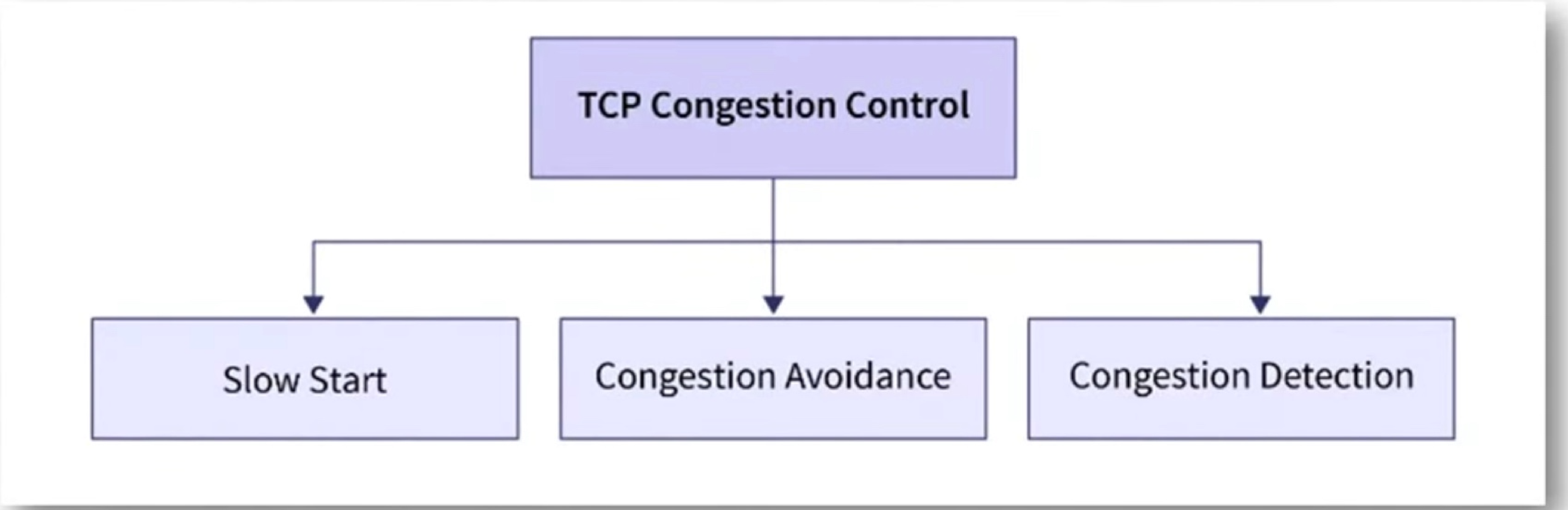
What is Congestion?

- Congestion happens when too many data packets are sent through a network which overwhelming the routers and switches that direct traffic.
- It increase packet delays, packet loss, wasting network recourses & degrade the throughput.
- TCP congestion control is a method used by the TCP protocol to manage data flow over a network and prevent congestion.
- TCP uses a congestion window and congestion policy that avoids congestion





TCP Congestion Policy / Approaches



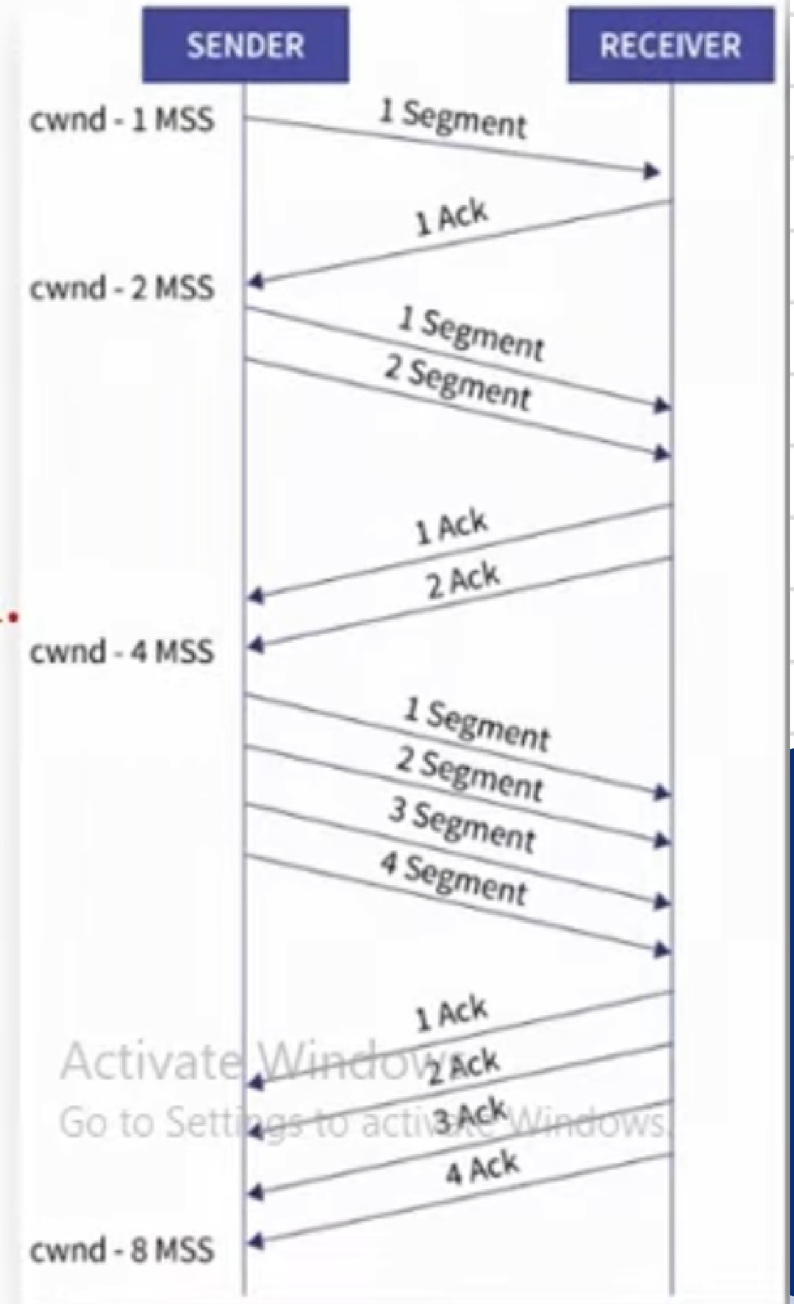
1. Slow Start Phase

- Initially, sender sets Congestion Window size = Maximum Segment Size (1 MSS)(e.g, 1 KB).
- After each successful acknowledgment from client, it doubles window size: 1 to 2, 2 to 4 & so on.
- This exponential growth continues until congestion window reaches a threshold (called the slow-start threshold) or packet loss is detected.

✓After 1 round trip time, congestion window size = (2)1 = 2 MSS

✓After 2 round trip time, congestion window size = (2)2 = 4 MSS

✓After 3 round trip time, congestion window size = (2)3 = 8 MSS and so on.



Threshold:

= Maximum number of TCP segments that receiver window can accommodate / 2

= (Receiver window size / Maximum Segment Size) / 2

Slow Start

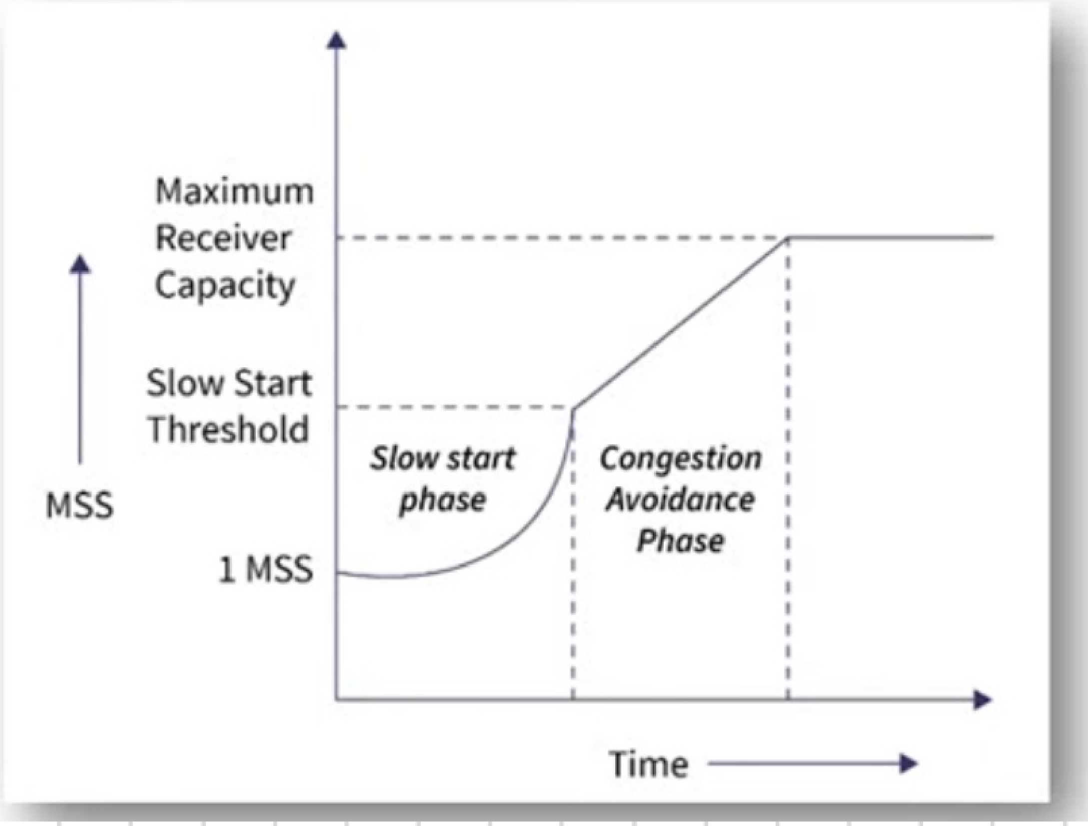
- Slow Start is one of the core algorithms used by
- Transmission Control Protocol to control congestion when a connection begins or after packet loss.
- Despite the name, it actually grows the sending rate very quickly at first—the "slow" part refers to starting cautiously from a small amount.
- When a connection starts, the sender has no idea about available network capacity.
- If it sends too much data immediately, it may cause congestion and packet loss.
- So TCP starts small and probes the network capacity gradually.

Slow Start

1. Congestion Window (cwnd)
    a. Limits how much data the sender can send without receiving ACKs.
    b. fteasured in segments (packets).
1. Slow Start Threshold (ssthresh)
    a. A limit that decides when to switch from Slow Start → Congestion Avoidance.

2. Congestion Avoidance Phase

- Suppose the threshold (ssthresh) is 16 segments.
- Once the congestion window reaches 16, the server stops doubling.
- And increases window linearly by adding one segment for each successful round-trip.
- This steady growth prevents the server from overwhelming the network.
- **Formula:** Congestion window size = Congestion window size + 1
- This phase continues until size of window becomes equal to that of the receiver window size.



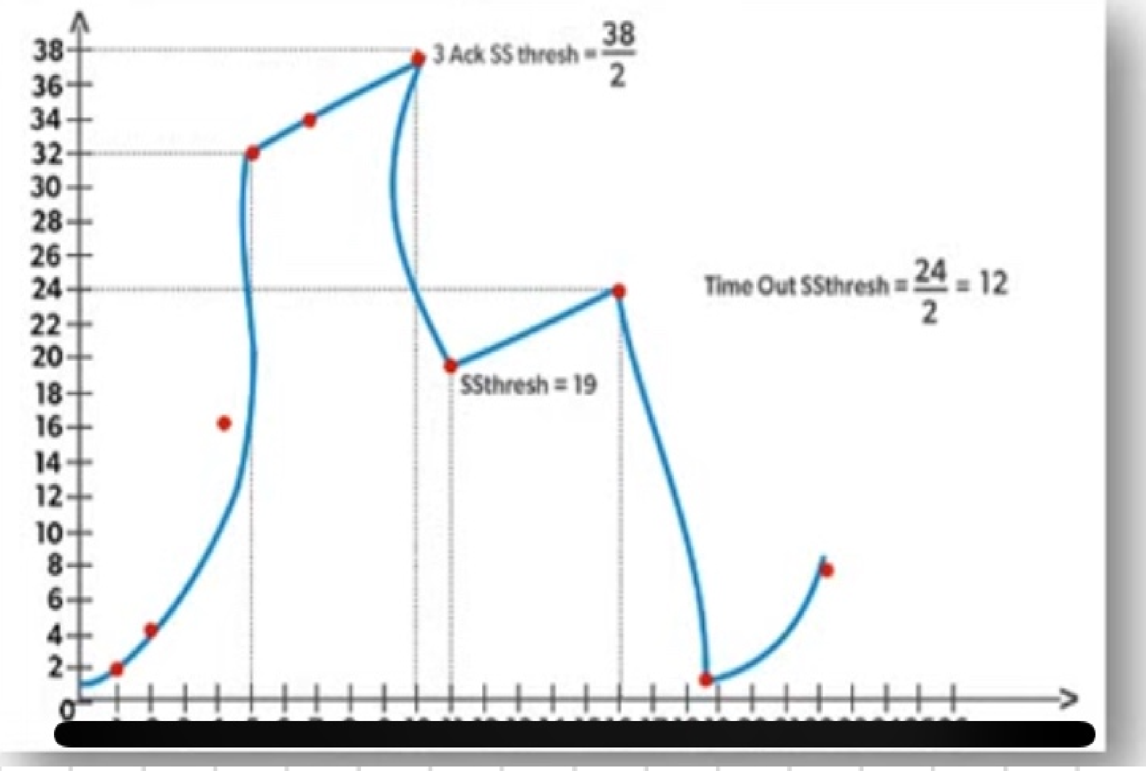
AIMD (Additive Increase, Multiplicative Decrease)

- AIftD is the fundamental strategy used by
- Transmission Control Protocol to control congestion and share network bandwidth fairly among multiple connections.
- It defines how the congestion window (cwnd) changes over time depending on network conditions.

3. Congestion Detection Phase

- If the network is congested or congestion window grows too large then packets might get dropped or loss.
- The server detects packet loss through Three duplicate ACKs or Timeout then it immediately reduces the congestion window as half the current size.

- **Example**:
- If packet loss happens at a congestion window of 24 then window size is reduced to 12.
- It slow start resumes until it reaches the congestion avoidance threshold.



AIMD (Additive Increase, Multiplicative Decrease)

Additive Increase

- Gradually increase cwnd by a small amount over time.
- Typically: +1 ftSS (maximum segment size) per RTT (Round Trip Time)
- This ensures slow, steady growth to avoid congestion.

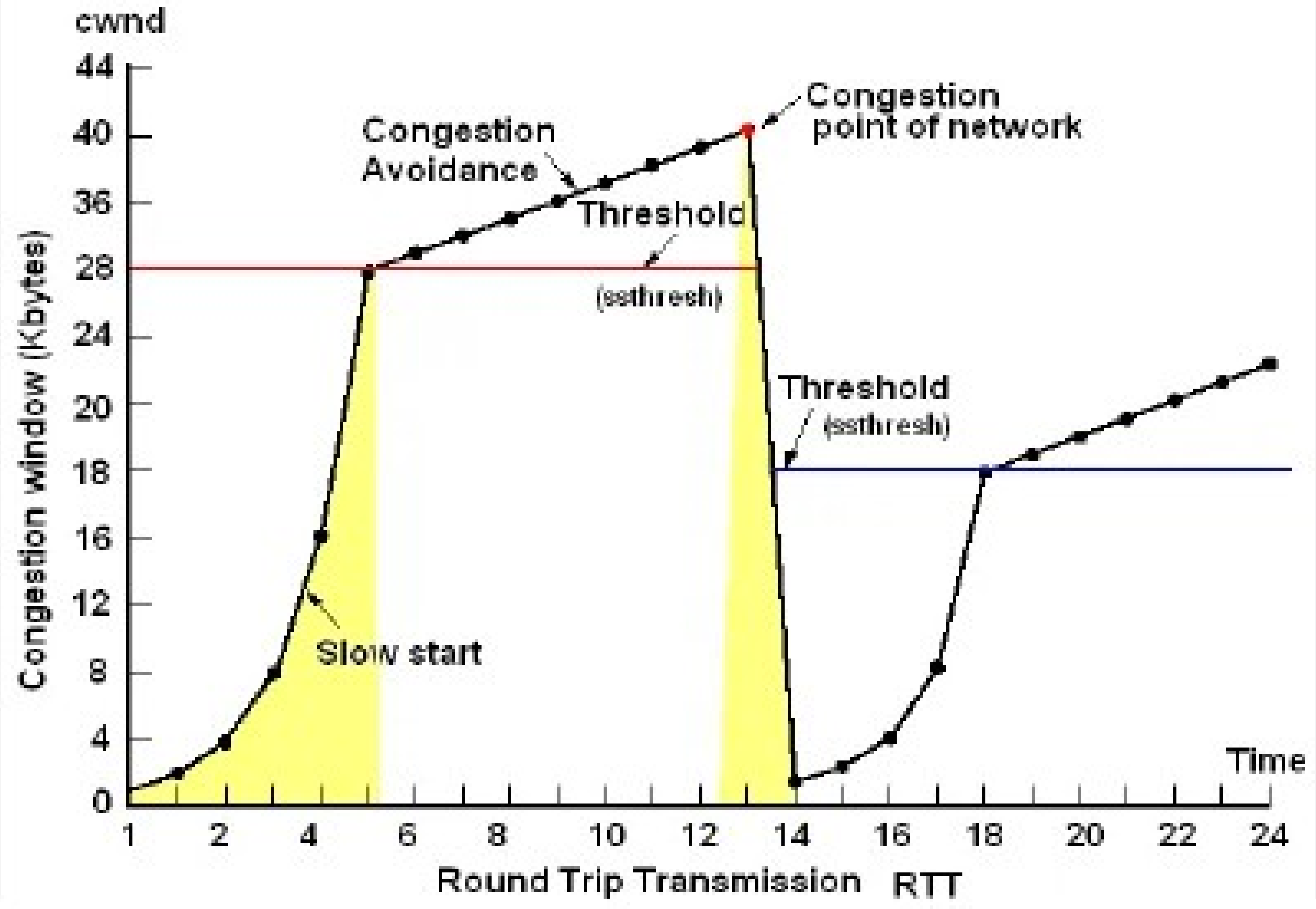
Multiplicative Decrease

- When congestion is detected (packet loss or ECN
- signal), cwnd is reduced sharply.
- Typically: cwnd = cwnd / 2

AIMD (Additive Increase, Multiplicative Decrease)

AIMD balances Three goals:
✅ Efficient use of bandwidth
✅ Fairness among multiple users
✅ Stability (avoids congestion collapse)

Slow Start


cwnd
Congestion window (Kbytes)
44
40
36
28
24
20
18
16
12
8
4
0
Congestion
Avoidance
Congestion
point of network
Threshold
(ssthresh)
Threshold
(ssthresh)
Slow start
Time
1
2
4
6
8
10
12
14
16
18
20
22
24
Round Trip Transmission RTT

AIMD (Additive Increase, Multiplicative Decrease)